

AGROSPEAR CROPCARE CO., LTD.

INSECTICIDE MANUFACTURING QUALITY CONTROL SYSTEM & DOSSIER

杀虫剂制剂生产全流程质量控制体系与佐证报告规程

Document Code: SOP-QC-INS-2026

Effective Date: August 2026

Version: Rev 4.0 (Global Standard)

Compliance: FAO/WHO Specs & CIPAC Methods

Product Lines: SC, EC, WDG, CS, ME Insecticides

1. END-TO-END INSECTICIDE QA/QC WORKFLOW / 杀虫剂生产品控全生命周期流程

QC Stage / 品控阶段	Key Control Parameters & Protocols / 关键管控项目与规范	Methodology / 检验方法与仪器	Release Criteria / 放行准则
IQC (Incoming QC) 原辅料进厂检验	• Technical Material (TC): AI purity, relevant toxic impurities, moisture, acidity. • 原药(TC): 有效成分纯度、异构体比例、有害杂质限度、水分及酸碱度。 • Solvents/Surfactants: Moisture, emulsification index, flash point, compatibility. • 溶剂/助剂/载体: 纯度、含水量、乳化力、闪点与共溶稳定性。 • Packaging: Fluorinated HDPE drop test, torque, barrier & foil seal test. • 包材: 氟化瓶/桶耐压防渗漏、跌落试验及铝箔封口相容性。	• HPLC-DAD / GC-FID • Karl Fischer Titrator • Drop & Vacuum Tester	100% Passed 不合格严禁入库投料
IPQC (In-Process) 制程过程巡检	• Emulsifiable Conc. (EC): Dissolution end-point, water content < 0.5%, cold turbidity. • 乳油(EC): 原药完全溶解度、微量水分(<0.5%)及 0°C 低温结晶监控。 • Suspension Conc. (SC): Milling bead temp < 38°C, fineness D90 < 4.0µm, structuring viscosity. • 悬浮剂(SC): 砂磨腔温升(<38°C)、激光粒径分布及触变沉降胶体构建。 • Microcapsule (CS): Interfacial polymerization temp, core-shell encapsulation efficiency > 98%. • 微囊悬浮剂(CS): 界面聚合反应温控、游离原药残留量与包封率(>98%)。 • Filling Line: Net volume accuracy (±0.5%), induction foil seal, laser batch code. • 灌装线: 每 30 分钟净含量抽检、高频电磁封口密封性与溯源防伪喷码。	• Laser Diffraction (D50/D90) • Brookfield Viscometer • Online Density Sensor • Dynamic Core Analysis	CCP In-Spec 偏离立即停线调整
FQC / OQC 成品终检与放行	• Active Ingredient Assay (AI % w/w or g/L) against FAO tolerances. • 标称有效成分含量精确测定 (严格符合 FAO 允许允差范围)。 • Physicochemical: Suspensibility, Emulsion stability, Persistent foam, pH, Wet sieving. • 理化指标: 悬浮率、乳化稳定性(标准水A/D)、起泡量、倾倒残余及湿筛试验。 • Accelerated Heat & Cold Storage: 54±2°C (14 days) and 0±2°C (7 days) stability validation. • 稳定性加速验证: 54°C 热储 14 天原药分解率 ≤ 5% 及 0°C 冷储无分层沉淀。	• CIPAC MT 184 / MT 36.3 • CIPAC MT 47.2 / MT 46.3 • Calibrated HPLC/GC	COA Approved QA 总监签署放行
Traceability & Retention 留样与质量追溯	• Retained samples stored in climate chamber for shelf-life + 1 year (min. 3 years). • 成品与原药留样存放于阴凉恒温留样室, 留存至保质期后 1 年 (至少 3 年)。 • 2D QR-code serialization for complete farm-to-factory dual-direction track and trace. • 一物一码赋码追溯系统, 实现原药批次、投料工序、机台与发货去向全流程追溯。	• Stability Chambers • Cloud Traceability ERP	Archived 建立完整批档案

2. FORMULATION-SPECIFIC QC STANDARDS (FAO/CIPAC COMPLIANT) / 主要杀虫剂剂型质控标准

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
SC (Suspension Conc.) 悬浮剂	• Chlorantraniliprole 200g/L SC • Thiamethoxam 30% SC • Lufenuron 50g/L SC	• Active Content: Nominal ± 5.0% • Particle Size: D50 ≤ 2.0µm, D90 ≤ 4.5µm • Suspensibility: ≥ 92.0% (after 54°C heat test) • Pourability Residue: ≤ 3.5%, Rinsed ≤ 0.5%	• CIPAC MT 184 • CIPAC MT 148.1 • Laser Diffraction
EC (Emulsifiable Conc.) 乳油	• Emamectin Benzoate 5% EC • Lambda-cyhalothrin 5% EC • Abamectin 1.8% / 5% EC	• Active Content: Nominal ± 5.0% • Water Content: ≤ 0.40% (prevent hydrolysis) • Emulsion Stability: 0.5h/2h/24h no cream/oil (<1mL) • Cold Stability: 0±2°C for 7 days (clear, no crystal)	• CIPAC MT 36.3 • CIPAC MT 39.3 • Karl Fischer
WDG / SG (Granules) 水分散粒剂 / 可溶粒剂	• Emamectin Benzoate 5% WDG • Acetamiprid 20% / 70% WDG • Dinotefuran 20% WDG	• Disintegration Time: ≤ 60 seconds • Suspensibility: ≥ 88.0% • Residual Moisture: ≤ 1.8% • Attrition Resistance: ≥ 95.0% (CIPAC MT 178.2)	• CIPAC MT 174 • CIPAC MT 185 • CIPAC MT 171

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
CS (Microcapsule Susp.) 微囊悬浮剂	• Lambda-cyhalothrin 10% CS • Chlorpyrifos 250g/L CS	• Free Active Ingredient (Unencapsulated): < 2.0% • Encapsulation Efficiency: ≥ 98.0% • Slow-Release Rate Profile (0h, 6h, 24h, 72h)	• Extraction HPLC • CIPAC MT 190

3. SUPPORTING EVIDENCE 1: OFFICIAL COA TEMPLATE / 佐证文件一：成品出厂检验报告单 (COA)

AGROSPEAR CROPCARE CO., LTD.
CERTIFICATE OF ANALYSIS / 产品检验分析报告单 (INSECTICIDE)

Product Name / 品名: Chlorantraniliprole 200g/L SC (氯虫苯甲酰胺 200g/L 悬浮剂)

Batch No. / 生产批号: AS-INS-20260826-03

Invoice / Order No. / 订单号: AGR-EXP-2026-889

Batch Quantity / 批量: 12,000 Liters (1L x 12/Carton)

Manufacture Date / 生产日期: 2026-08-26

Expiry Date / 有效期至: 2028-08-25

Sampling Date / 取样日期: 2026-08-26

Inspection Date / 判定日期: 2026-08-27

Test Parameter / 检验项目	Specification Limit / 标准指标	Test Result / 实测结果	Standard Method / 检验方法	Verdict / 判定
Appearance / 外观性状	Off-white homogeneous flowable suspension (类白色均质流体)	Conforms / 符合要求	Visual Check	Pass
Active Ingredient / 有效成分含量	200.0 ± 10.0 g/L (190.0 – 210.0 g/L)	201.8 g/L	HPLC (Agilent 1260)	Pass
pH Value (1% aqueous solution)	5.5 – 8.5	6.7	CIPAC MT 75.3	Pass
Suspensibility / 悬浮率	≥ 90.0 % (CIPAC Water D)	96.4 %	CIPAC MT 184	Pass
Particle Size Distribution / 粒度 (D90)	≤ 4.5 µm	2.9 µm (D50: 1.4 µm)	Laser Diffraction	Pass
Wet Sieve Test / 湿筛试验 (75µm sieve)	≥ 99.0 % through	99.9 %	CIPAC MT 185	Pass
Persistent Foaming / 持久起泡性	≤ 40 mL after 1 min	12 mL	CIPAC MT 47.2	Pass
Pourability / 倾倒性 (Rinsed Residue)	Residue ≤ 4.0 %, Rinsed ≤ 0.5 %	Residue: 1.8 %, Rinsed: 0.2 %	CIPAC MT 148.1	Pass
Heat Stability (54±2°C, 14 days) / 热储	AI Degradation ≤ 5.0%, Suspensibility ≥ 90%	Degradation: 0.9%, Susp: 95.1%	CIPAC MT 46.3	Pass
Low Temp. Stability (0±2°C, 7 days) / 冷储	No crystal growth or phase separation (无晶体析出)	Conforms / 无析出	CIPAC MT 39.3	Pass

CONCLUSION / 检验结论: PASS - All parameters conform to FAO Specification and In-house Quality Standard. Approved for release. (合格 - 各项指标符合 FAO 规格及企业内控标准, 准予放行出厂。)

Analyst / 检测员:
Zhang Wei (张伟)
Date: 2026-08-27Reviewer / 复核员:
Li Ming (李明)
Date: 2026-08-27QA Director / QA总监:
Dr. Alex Chen (陈博士) [Approved]
Date: 2026-08-27

4. SUPPORTING EVIDENCE 2: BATCH RELEASE & OOS DISPOSITION / 佐证文件二：批质量放行审批与超标处理规程

Release Gate / 质量关卡	Verification Requirements / 审核项目	Status / 状态	Verified By / 审核人
1. Raw Material Gate (IQC)	Active ingredient purity, solvent flash point, container foil compatibility	Verified Pass	IQC Lead
2. Process Parameters (IPQC)	Milling chamber temp < 38°C, filtration integrity, torque & volume accuracy	Verified Pass	IPQC Officer
3. Laboratory FQC (COA)	Active assay, HPLC chromatogram purity, suspensibility, cold/heat stability	Verified Pass	Lab Supervisor
4. Packaging & Serialization	Dual 2D QR-code serialization, batch number & expiry laser coding legibility	Verified Pass	Packaging QA

OOS (Out of Specification) & CAPA Protocol: If any batch fails specifications, the Batch Release Gate is locked automatically in the ERP. Investigation is conducted across 3 tiers (Analytical error → Formulation process deviation → Raw material drift). Reprocessing requires a formal CAPA protocol approved by the Plant General Manager and QA Director.

超标调查 (OOS) 与纠正预防措施 (CAPA): 凡任一指标出现偏差, ERP 系统自动锁定该批产品出厂闸口并张贴物理红卡。启动三级溯源调查 (实验室分析误差 → 生产工艺偏离 → 原料批次漂移)。返工或重组方案必须经工厂总经理与质量总监双重书面签批方可执行。